

CANDO, ND, USA

WMO: 720858

Lat: 48.480N

Lon: 99.236W

Elev: 450

StdP: 96.03

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 00285

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.9	-27.6	-33.0	0.2	-29.6	-31.1	0.2	-27.4	14.3	-10.4	12.8	-12.7	3.4	290	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.0	20.9	28.0	20.2	27.3	19.6	23.5	27.0	22.3	26.1	21.1	25.1	5.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.2	25.0	21.2	16.7	24.0	19.9	15.4	23.3	72.4	27.1	67.5	25.2	63.3	24.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.2	10.0	-33.0	34.0	2.2	2.1	-34.6	35.5	-35.9	36.7	-37.1	37.9	-38.7	39.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-13.5	-14.3	-5.1	2.7	11.7	17.3	19.7	18.5	13.6	5.0	-4.0	-11.2		
	DBStd	13.51	7.83	7.10	8.00	5.82	4.98	3.13	2.80	3.14	4.51	4.77	6.41	7.24		
	HDD10.0	3401	729	679	468	226	39	1	0	0	16	167	419	657		
	HDD18.3	5571	987	913	726	470	214	54	18	38	153	413	669	916		
	CDD10.0	1019	0	0	2	5	91	220	301	263	124	12	1	0		
	CDD18.3	145	0	0	0	0	7	24	61	43	11	0	0	0		
	CDH23.3	1654	0	0	0	2	145	250	581	516	157	3	0	0		
	CDH26.7	406	0	0	0	0	39	41	131	154	41	0	0	0		
(14) Wind	WSAvg	4.8	5.3	5.0	4.9	5.4	5.3	4.7	3.8	3.6	4.4	5.1	4.9	4.9		
(15) Precipitation	PrecAvg	492	11	12	21	31	70	93	84	63	47	37	21	15		
	PrecMax	631	33	50	53	67	173	200	237	172	116	118	79	31		
	PrecMin	310	1	0	4	3	22	31	29	26	8	1	0	4		
	PrecStd	93	8	11	14	17	39	42	47	33	28	30	23	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	2.8	17.5	22.5	30.9	29.7	31.4	33.6	30.2	22.8	14.9	6.1		
		MCWB	1.8	1.6	11.9	11.6	17.4	21.0	22.8	22.1	19.8	14.2	9.8	2.3		
	2%	DB	2.2	0.9	12.8	18.0	27.4	27.6	29.1	30.1	27.5	18.9	9.8	2.5		
		MCWB	0.9	0.3	8.9	9.9	16.4	19.1	22.4	21.0	18.1	11.4	6.8	0.8		
	5%	DB	0.1	-1.0	8.7	15.1	24.1	26.3	27.8	27.9	24.9	16.0	7.0	0.1		
		MCWB	-0.5	-1.5	6.4	8.7	13.6	18.5	21.5	20.0	16.7	10.7	4.4	-0.3		
	10%	DB	-2.1	-2.8	4.9	12.5	22.1	24.7	27.1	26.3	22.4	12.9	3.9	-2.1		
		MCWB	-2.6	-3.4	3.2	7.2	13.3	17.8	20.7	19.3	16.0	8.9	2.5	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	1.8	13.1	12.7	19.4	23.7	26.4	24.1	22.8	16.0	10.2	3.3		
		MCDB	3.6	2.9	15.6	20.1	25.1	26.5	29.4	28.6	26.2	19.3	13.9	5.1		
	2%	WB	0.8	0.2	9.0	11.0	18.1	21.9	24.1	22.6	20.6	13.0	7.1	1.1		
		MCDB	1.8	0.6	12.4	16.6	23.3	24.9	27.1	27.6	24.0	16.1	9.7	2.4		
	5%	WB	-0.6	-1.7	6.0	9.2	16.3	20.3	22.8	21.2	18.7	10.9	4.5	-0.3		
		MCDB	-0.1	-1.2	8.7	14.8	21.8	23.9	26.0	26.1	23.5	15.3	6.3	0.3		
	10%	WB	-2.4	-3.6	3.2	7.7	14.5	19.0	21.7	20.1	17.0	9.3	2.7	-2.2		
		MCDB	-2.2	-2.8	4.7	11.6	19.1	23.0	25.6	24.8	20.9	13.4	3.9	-1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	10.6	10.3	11.1	12.8	11.2	11.9	13.7	13.1	11.0	9.1	8.8		
		MCDBR	11.5	10.8	14.0	17.2	17.7	13.9	13.4	16.0	16.8	16.6	12.8	10.7		
		MCWBR	10.6	10.0	9.7	9.6	8.2	7.3	7.2	8.2	8.6	9.2	9.3	9.1		
		MCDBR	11.5	10.4	13.5	16.2	14.2	11.3	12.0	14.3	15.0	14.4	12.5	10.5		
	5% WB	MCWBR	10.7	9.7	9.6	9.5	7.5	7.7	7.7	8.1	8.6	8.8	9.3	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.285	0.295	0.355	0.368	0.373	0.380	0.381	0.347	0.295	0.266	0.246		
	taud		2.277	2.283	2.429	2.327	2.350	2.385	2.371	2.353	2.428	2.552	2.493	2.428		
	Ebn at Noon		787	872	929	894	893	888	875	857	857	863	818	801		
	Edh at Noon		73	92	94	116	119	116	116	113	95	70	59	55		
(44) All-Sky Solar Radiation	RadAvg		1.60	2.76	3.96	4.97	5.64	6.09	6.46	5.32	3.92	2.45	1.45	1.11		
	RadStd		0.11	0.25	0.34	0.40	0.36	0.41	0.33	0.35	0.26	0.30	0.20	0.13		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page